Yang–Mills Dynamics from Projective Spectral Entropy: Vertical Heat-Kernel Variation on the Admissible Fibre

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Abstract

The companion paper [6] showed that the horizontal variation of the projective spectral entropy functional $S_\Pi[g] = \frac{1}{2} \log \det' A_g$ with respect to the base metric produces the Einstein tensor as the infrared-dominant response, via the Seeley–DeWitt coefficient a_2. The present paper carries out the complementary vertical variation: extending A_g to a Laplace-type operator $A_{g,\mathcal{A}} = -(\nabla^{\mathcal{A}})^2 + E$ on the associated vector bundle of the admissible principal fibre, we compute the Seeley–DeWitt coefficient a_4 for the total operator and extract the Yang–Mills term $c_{\mathrm{YM}} \int \mathrm{Tr}(F_{\mu\nu} F^{\mu\nu}) \sqrt{g}\, \mathrm{d}^4 x$. Variation of $S_\Pi[g, \mathcal{A}]$ with respect to the admissible connection $\mathcal{A}$ then yields the Yang–Mills equations $D_\mu F^{a\mu\nu} = 0$ in the current-free sector. The induced gauge coupling $g_{\mathrm{YM}}^{-2} \sim (12 \cdot 16\pi^2)^{-1} \log(\Lambda/\mu)$ is logarithmic in the UV cutoff, while Newton's constant $G_N^{-1} \sim \ell_{\mathrm{sp}}^{-2}$ is quadratic. This spectral hierarchy is a structural consequence of the Seeley–DeWitt expansion and does not require a separate input. The gauge group G_Π is taken as input from the spectral admissibility programme [5, 8]; the SU(3) sector is unconditional at the effective level: $[H\text{-color}]_{\mathrm{eff}}$ (equality of capacity exponents $\delta_c = \delta_{\omega c} = \delta_{\omega^2 c}$) is established analytically in [7, 8]. The central message is: $a_2 \to$ Einstein, $a_4 \to$ Yang–Mills, from the single functional $S_\Pi[g, \mathcal{A}]$. This spectral stratification suggests that the natural organisational space for unification is spectral and variational rather than purely group-theoretic: gravity and gauge dynamics are separated by geometric direction and heat-kernel order, not by a missing symmetry group.

Keywords: Yang–Mills equations, heat-kernel expansion, Seeley–DeWitt coefficients, spectral entropy, induced gauge theory, gauge–gravity hierarchy, admissible fibre, projective non-injectivity.

Contents

1 Introduction

The Cosmochrony programme derives physical structure from a static relational substrate χ via a generally non-injective projection Π [1, 3]. Two lines of development have reached sufficient maturity to be combined. The first, pursued in the Gravity paper [6] and completed in Gravity 3.0, shows that the horizontal variation of the projective spectral entropy functional

$$S_\Pi[g] \;=\; \tfrac{1}{2}\log \det{}' A_g \tag{1}$$

with respect to the base metric $g_{\mu\nu}$ reproduces the Einstein–Hilbert term as the infrared-dominant contribution [6]. The second, pursued in the Q-series from Q6a onward, derives the admissible gauge group G_Π, gauge charges, and the Yang–Mills principle from the fibre structure of Π [5].

These two results share a common object, A_g, but exploit it in orthogonal directions. The present paper unifies them by extending A_g to a Laplace-type operator $A_{g,\mathcal{A}}$ on the total associated bundle of Π and computing both the gravitational (a_2) and the gauge (a_4) sectors of the spectral entropy from a single functional.

1.1 Main result

The central result is the following.

Theorem 1 (Yang–Mills from a_4)**.** *Let G_Π be the admissible gauge group derived in [5, 8], and let $A_{g,\mathcal{A}}$ be the Laplace-type operator on the associated vector bundle constructed in Section 3. Then:*

(i) *The Seeley–DeWitt coefficient $a_4(A_{g,\mathcal{A}})$ contains the Yang–Mills density:*

$$a_4(A_{g,\mathcal{A}}) \;\supset\; \frac{1}{16\pi^2}\int \frac{1}{12}\,\mathrm{Tr}(\Omega_{\mu\nu}\Omega^{\mu\nu})\sqrt{g}\,\mathrm{d}^4x\,, \quad \Omega_{\mu\nu} = F_{\mu\nu}\,.$$

(ii) *The projective spectral entropy contains the renormalized Yang–Mills term:*

$$S_\Pi[g,\mathcal{A}] \;\supset\; \frac{1}{12\cdot 16\pi^2}\,\log\!\left(\frac{\Lambda}{\mu}\right)\int \mathrm{Tr}(F_{\mu\nu}F^{\mu\nu})\sqrt{g}\,\mathrm{d}^4x\,.$$

(iii) *The vertical variation of S_Π yields the Yang–Mills equations in the current-free sector:*

$$\frac{\delta S_\Pi}{\delta \mathcal{A}^a_\nu} \;=\; 0 \quad\Longrightarrow\quad D_\mu F^{a\mu\nu} = 0\,.$$

These three statements follow from standard heat-kernel technology [10, 12] applied to the operator $A_{g,\mathcal{A}}$, once the base–fibre structure and the admissibility of $\mathcal{A}$ are established.

1.2 Position in the programme

The logical chain of the present paper is:

$$\Pi \;\Rightarrow\; P_{G_\Pi}(M, G_\Pi) \;\Rightarrow\; \nabla^{\mathcal{A}} \;\Rightarrow\; A_{g,\mathcal{A}} = -(\nabla^{\mathcal{A}})^2 + E \;\Rightarrow\; a_4 \supset \tfrac{1}{12}\,\mathrm{Tr}(F^2) \;\Rightarrow\; D_\mu F^{\mu\nu} = 0\,.$$

The companion paper [6] established the horizontal half of the same chain:

$$A_g \;\Rightarrow\; a_2 \supset R \;\Rightarrow\; G_{\mu\nu} = 0\,.$$

Together, they show that the functional $S_\Pi[g, \mathcal{A}]$ produces Einstein gravity and Yang–Mills gauge dynamics as the a_2 and a_4 sectors of a single spectral entropy expansion.

The gauge-gravity synthesis — including the joint equations of motion, the ratio $G_N g_{\mathrm{YM}}^2$, and the Eddington–Born–Infeld completion of the gauge sector — is deferred to the companion paper Q13.

1.3 Organisation

Section 2 establishes the base–fibre decomposition of the projective operator and proves the horizontal–vertical decoupling lemma. Section 3 constructs $A_{g,\mathcal{A}}$ on the associated bundle. Section 4 computes the a_4 coefficient and identifies the Yang–Mills term. Section 5 carries out the vertical variation and derives the Yang–Mills equations. Section 6 establishes the induced coupling and the gauge–gravity spectral hierarchy. Section 7 collects the epistemic status of each result and the analytical status of [H-color]. Section 8 concludes.

2 Base–fibre decomposition of the projective operator

2.1 The admissible principal bundle

The non-injective projection $\Pi : \Omega \to O$ of Foundation M [1] carries a natural base–fibre structure. The effective base space M is the Riemannian manifold derived in Q5b [4] and Q11 [9]: the BFS shell stratification of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ converges in the Gromov–Hausdorff sense to a Carnot–Carathéodory structure with homogeneous dimension $D_{\mathrm{hom}} = 4$ whose effective metric closure is Lorentzian, as established in [9]. The effective fibre is the admissible gauge group $G_\Pi = \mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ derived in [5, 8] as the composite of three admissibility fixed points of orbit sizes 1, 2, and 3.

Definition 1 (Admissible principal bundle)**.** The admissible principal bundle associated with the projection Π is the principal G_Π-bundle $\pi : P \to M$, with structure group G_Π and connection $\mathcal{A}$, constructed as follows. The total space P is the fibre product of the three orbit sub-bundles identified in [5, 8]. The connection $\mathcal{A} \in \Omega^1(P, \mathfrak{g}_\Pi)$ is the unique ad-equivariant horizontal distribution compatible with the admissibility constraints of Foundation M [1]. The curvature $F_{\mu\nu} = \partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu + [\mathcal{A}_\mu, \mathcal{A}_\nu]$ takes values in $\mathfrak{g}_\Pi$.

2.2 The associated vector bundle

Let $\rho : G_\Pi \to \mathrm{GL}(V)$ be a representation of G_Π on a finite-dimensional complex vector space V. The associated vector bundle is $\mathcal{V} = P \times_\rho V \to M$. Sections of $\mathcal{V}$ carry the combined action of the base geometry (through the Levi-Civita connection ∇^g) and the fibre gauge field (through $\mathcal{A}$). The covariant derivative on $\Gamma(\mathcal{V})$ is

$$\nabla^{\mathcal{A}}_\mu \;=\; \partial_\mu + \omega_\mu + \mathcal{A}_\mu \,, \tag{2}$$

where ω_μ is the spin connection on M.

2.3 Horizontal–vertical decoupling

The following lemma is essential for the independence of the gravitational (a_2) and Yang–Mills (a_4) sectors.

Lemma 1 (Horizontal–vertical decoupling at leading order)**.** *At fixed admissible projection fibre, variations of the base metric g and of the fibre connection $\mathcal{A}$ define independent horizontal and vertical variations of the total Laplace-type operator $A_{g,\mathcal{A}}$. Mixed variations $\delta^2 A_{g,\mathcal{A}}/\delta g\,\delta\mathcal{A}$ contribute only at order a_6 and higher in the Seeley–DeWitt expansion and do not modify the leading a_2 Einstein sector or the leading a_4 Yang–Mills sector.*

Proof. The Seeley–DeWitt coefficients $a_{2k}(A_{g,\mathcal{A}})$ are local integrals of invariants of degree $2k$ in derivatives of $g_{\mu\nu}$ and $\mathcal{A}_\mu$ [12]. At minimal coupling, the leading gauge-dependent contribution to a_4 is the pure Yang–Mills invariant $\mathrm{Tr}(\Omega_{\mu\nu}\Omega^{\mu\nu})$. Mixed gauge–gravity terms of the form $R\,\mathrm{Tr}(F^2)$ are either absent at this order or reduce to curvature renormalizations that are independent of the vertical variation $\delta_{\mathcal{A}}$ and therefore do not affect the Yang–Mills sector. The a_2 coefficient depends only on g and E (not on $\mathcal{A}$ at minimal coupling), and the a_4 coefficient thus splits cleanly into a purely gravitational part and a purely gauge part. Horizontal and vertical variations are therefore decoupled at the a_2 and a_4 levels. $\square$

Remark. The decoupling holds under minimal coupling, i.e. when E does not depend on $\mathcal{A}$. Curvature couplings of the form $E \supset \kappa\, F_{\mu\nu} g^{\mu\rho} g^{\nu\sigma}$ would introduce mixed a_4 terms; the admissibility constraints of Foundation M [1] impose minimal coupling as the natural choice at leading order.

3 Laplace-type operator with admissible fibre connection

3.1 Construction of the operator

The spectral entropy functional of the Gravity paper [6] is extended to the associated bundle $\mathcal{V}$ by replacing A_g with the operator:

$$A_{g,\mathcal{A}} \;=\; -(\nabla^{\mathcal{A}})^2 + E\,, \tag{3}$$

acting on $L^2(M, \mathcal{V})$. Here $(\nabla^{\mathcal{A}})^2 = g^{\mu\nu}\nabla^{\mathcal{A}}_\mu \nabla^{\mathcal{A}}_\nu$ is the connection Laplacian, and $E \in \mathrm{End}(\mathcal{V})$ is the endomorphism encoding the admissibility correction:

$$E \;=\; \frac{R}{6}\,\mathrm{id}_{\mathcal{V}} + E_{\mathrm{adm}}\,, \tag{4}$$

where $R/6\,\mathrm{id}_{\mathcal{V}}$ is the Lichnerowicz–Weitzenböck correction ensuring minimal coupling to the scalar curvature, and E_{adm} encodes the admissibility constraints of the projection fibre.

3.2 Self-adjointness and non-negativity

Proposition 1. *The operator $A_{g,\mathcal{A}}$ is self-adjoint and non-negative on $L^2(M, \mathcal{V})$ for any admissible connection $\mathcal{A}$ and any $E \geq 0$.*

Proof. The connection Laplacian $-(\nabla^{\mathcal{A}})^2$ is self-adjoint and non-negative by standard functional analysis [10]. The endomorphism E is self-adjoint by construction (the Lichnerowicz term is a scalar multiple of the identity, and E_{adm} is symmetric under the admissibility constraints of [1]). Non-negativity for $E \geq 0$ follows from the Bochner identity. □

3.3 The extended spectral entropy

The spectral entropy functional is extended to:

$$S_\Pi[g, \mathcal{A}] \;=\; \tfrac{1}{2}\,\log \mathrm{det}' A_{g,\mathcal{A}}\,, \tag{5}$$

where $\det'$ denotes the zeta-regularized determinant excluding the zero modes. When $\mathcal{A} = 0$ and $E = R/6\,\mathrm{id}$, this reduces to the functional of [6].

4 Heat-kernel coefficient a_4 and the Yang–Mills term

4.1 Seeley–DeWitt expansion

For the operator $A_{g,\mathcal{A}}$ of Equation (3), the heat-kernel expansion as $t \to 0^+$ takes the standard form [10, 12]:

$$\mathrm{Tr}\big(e^{-t\,A_{g,\mathcal{A}}}\big) \;\sim\; (4\pi t)^{-2} \sum_{k=0}^{\infty} t^k\, a_{2k}(A_{g,\mathcal{A}})\,, \quad t \to 0^+\,, \tag{6}$$

where in four dimensions:

$$a_0 = \int \mathrm{tr}(\mathrm{id})\,\sqrt{g}\,\mathrm{d}^4x\,, \tag{7}$$

$$a_2 = \int \mathrm{tr}\left(\frac{R}{6}\,\mathrm{id} - E\right)\sqrt{g}\,\mathrm{d}^4x\,, \tag{8}$$

and a_4 is given in Section 4.3 below.

4.2 The a_2 coefficient and the Einstein sector

For $E = R/6\,\mathrm{id}_{\mathcal{V}} + E_{\mathrm{adm}}$, the a_2 coefficient reduces (at minimal coupling) to:

$$a_2(A_{g,\mathcal{A}}) \;=\; -\int \mathrm{tr}(E_{\mathrm{adm}})\,\sqrt{g}\,\mathrm{d}^4x\,, \tag{9}$$

which is independent of the gauge connection $\mathcal{A}$. Under the identification of the effective metric established in [6], the variation $\delta_g S_\Pi = \delta_g[\frac{1}{2}\log\det' A_{g,\mathcal{A}}]$ reproduces the Einstein tensor as the infrared-dominant contribution; this is the result of the Gravity paper [6] and is not repeated here. Lemma 1 guarantees that this sector is not modified by the gauge connection at orders a_2 and a_4.

4.3 The a_4 coefficient

The full a_4 coefficient for the operator $D = -(\nabla^{\mathcal{A}})^2 + E$ acting on sections of a vector bundle with fibre curvature $\Omega_{\mu\nu} = [\nabla^{\mathcal{A}}_\mu, \nabla^{\mathcal{A}}_\nu]$ is [12]:

$$a_4(D) = \frac{1}{16\pi^2}\int \mathrm{tr}\Big[\tfrac{1}{12}\,\Omega_{\mu\nu}\Omega^{\mu\nu} + \tfrac{1}{2}E^2 - \tfrac{1}{6}RE + \tfrac{1}{180}\big(5R^2 - 2R_{\mu\nu}R^{\mu\nu} + 2R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}\big)\,\mathrm{id} + \text{total derivative} \tag{10}$$

The terms involving R, $R_{\mu\nu}$, $R_{\mu\nu\rho\sigma}$ are purely gravitational and do not depend on $\mathcal{A}$. The term $\frac{1}{2}E^2$ contributes to the cosmological sector and to the admissibility renormalization. The gauge-dependent term is:

$$a_4(D)\big|_{\mathrm{gauge}} \;=\; \frac{1}{16\pi^2}\int \frac{1}{12}\,\mathrm{Tr}(\Omega_{\mu\nu}\Omega^{\mu\nu})\sqrt{g}\,\mathrm{d}^4x\,, \tag{11}$$

where Tr denotes the trace over the fibre representation ρ of G_Π.

4.4 Identification of the curvature two-form

The fibre curvature of the admissible connection $\mathcal{A}$ coincides with the gauge field strength:

$$\Omega_{\mu\nu} \;=\; [\nabla^{\mathcal{A}}_\mu, \nabla^{\mathcal{A}}_\nu] \;=\; \partial_\mu\mathcal{A}_\nu - \partial_\nu\mathcal{A}_\mu + [\mathcal{A}_\mu, \mathcal{A}_\nu] \;=\; F_{\mu\nu}\,. \tag{12}$$

This identification is standard for connection Laplacians on principal bundles and holds for any representation ρ of G_Π. Substituting into (11):

$$a_4(A_{g,\mathcal{A}})\big|_{\mathrm{gauge}} \;=\; \frac{1}{16\pi^2}\int \frac{1}{12}\,\mathrm{Tr}(F_{\mu\nu}F^{\mu\nu})\sqrt{g}\,\mathrm{d}^4x\,. \tag{13}$$

This establishes part (i) of Theorem 1.

5 Vertical variation and Yang–Mills equations

5.1 The spectral entropy and its renormalized form

The zeta-regularized spectral entropy (5) is related to the heat kernel by:

$$S_\Pi[g,\mathcal{A}] \;=\; -\tfrac{1}{2}\,\zeta'_{A_{g,\mathcal{A}}}(0)\,, \tag{14}$$

where $\zeta_D(s) = \mathrm{Tr}(D^{-s})$ is the spectral zeta function. After UV regularization with cutoff Λ and renormalization scale μ, the gauge-dependent part of the entropy acquires the contribution:

$$S_\Pi[g,\mathcal{A}] \;\supset\; \frac{1}{12\cdot 16\pi^2}\,\log\!\left(\frac{\Lambda}{\mu}\right)\int \mathrm{Tr}(F_{\mu\nu}F^{\mu\nu})\sqrt{g}\,\mathrm{d}^4x\,, \tag{15}$$

which establishes part (ii) of Theorem 1. The logarithmic form of the UV divergence is the hallmark of a renormalizable gauge sector in four dimensions.

5.2 Vertical variation

The variation of $S_\Pi[g,\mathcal{A}]$ with respect to the gauge connection $\mathcal{A}^a_\mu$ is obtained from (15) via:

$$\begin{aligned}\delta_{\mathcal{A}}\left[\mathrm{Tr}(F_{\mu\nu}F^{\mu\nu})\right] &= 2\,\mathrm{Tr}[F^{\mu\nu}\,\delta F_{\mu\nu}]\\ &= 2\,\mathrm{Tr}[F^{\mu\nu}(D_\mu\delta\mathcal{A}_\nu - D_\nu\delta\mathcal{A}_\mu)]\\ &= 4\,\mathrm{Tr}[F^{\mu\nu}D_\mu\delta\mathcal{A}_\nu]\;.\end{aligned} \tag{16}$$

Integrating by parts (using $\nabla_\mu\sqrt{g}=0$):

$$\int \mathrm{Tr}[F^{\mu\nu}D_\mu\delta\mathcal{A}_\nu]\,\sqrt{g}\,\mathrm{d}^4x = -\int \mathrm{Tr}[(D_\mu F^{\mu\nu})\delta\mathcal{A}_\nu]\,\sqrt{g}\,\mathrm{d}^4x + \text{boundary terms}\,. \tag{17}$$

Therefore:

$$\frac{\delta S_\Pi}{\delta\mathcal{A}^a_\nu} \;=\; -\frac{1}{3\cdot 16\pi^2}\,\log\!\left(\frac{\Lambda}{\mu}\right)\,(D_\mu F^{a\mu\nu})\,\sqrt{g}\,. \tag{18}$$

5.3 Yang–Mills equations

In the absence of projected matter currents (the "current-free sector"), the variational principle $\delta_{\mathcal{A}}S_\Pi = 0$ yields:

$$\boxed{D_\mu F^{a\mu\nu} = 0\,,} \tag{19}$$

which are the Yang–Mills equations for the gauge group G_Π in vacuo. This establishes part (iii) of Theorem 1.

Remark. When projected matter fields ψ are included, the variation of the matter coupling term in S_Π contributes a current $J^{a\nu} = \delta S_{\text{matter}}/\delta\mathcal{A}^a_\nu$, and the full equations become $D_\mu F^{a\mu\nu} = J^{a\nu}$. The derivation of the matter current from the projective structure lies beyond the scope of the present paper.

Remark. The Yang–Mills equations (19) emerge from S_Π via the a_4 coefficient, which is four orders of derivatives higher than the a_2 Einstein sector. This is not a deficiency but the correct spectral signature: the gauge sector is renormalizable (logarithmic coupling) while the gravitational sector is non-renormalizable (quadratic coupling). The Seeley–DeWitt expansion encodes this hierarchy automatically.

6 Induced gauge coupling and gauge–gravity hierarchy

6.1 Induced gauge coupling

Comparing the Yang–Mills term (15) with the standard Yang–Mills action $S_{\mathrm{YM}} = -(4g_{\mathrm{YM}}^2)^{-1} \int \mathrm{Tr}(F_{\mu\nu}F^{\mu\nu})\sqrt{g}\,\mathrm{d}^4x$ (in Euclidean signature), the induced coupling is:

$$g_{\mathrm{YM}}^{-2} \sim \frac{1}{12 \cdot 16\pi^2} \log\left(\frac{\Lambda}{\mu}\right) . \tag{20}$$

This logarithmic induction is characteristic of asymptotically free gauge theories in four dimensions and is the standard result of the induced gauge theory programme [11].

6.2 Newton's constant from the same spectral cutoff

The induced Newton's constant from the companion paper [6] is:

$$G_N^{-1} \sim \frac{\dim V}{16\pi^2}\, \ell_{\mathrm{sp}}^{-2} , \tag{21}$$

where ℓ_{sp} is the spectral capacity length (the same cutoff scale that appears in the Born–Infeld completion of the gravitational sector [2, 6]). Here $\dim V = \sum_i \dim V_i$ is the total dimension of the representation bundle.

6.3 Gauge–gravity spectral hierarchy

The ratio of the two couplings is:

$$G_N\, g_{\mathrm{YM}}^2 \sim \frac{\ell_{\mathrm{sp}}^2}{\dim V} \cdot \log\left(\frac{\Lambda}{\mu}\right)^{-1} . \tag{22}$$

For $\Lambda \gg \mu$ (UV cutoff well above the renormalization scale) and $\ell_{\mathrm{sp}} \ll 1$ (spectral length below the Planck scale), this ratio satisfies $G_N g_{\mathrm{YM}}^2 \ll 1$, reproducing the observed hierarchy between gravitational and gauge strengths.

Remark. The hierarchy $G_N g_{\mathrm{YM}}^2 \ll 1$ is a structural consequence of the Seeley–DeWitt expansion and requires no separate fine-tuning. It follows from the fact that gravity is induced at order a_2 (quadratic UV divergence) while gauge dynamics are induced at order a_4 (logarithmic UV divergence). The two sectors share the same spectral cutoff ℓ_{sp}, which is the only scale in the problem beyond the renormalization group flow.

7 Epistemic status and relation to the Standard Model gauge group

7.1 Epistemic status of each result

Table 1 collects the epistemic status of the main results of this paper.

Table 1: Epistemic status of the main results.

Result	Status	Dependence
$a_4 \supset \frac{1}{12}\mathrm{Tr}(\Omega^2)$	Standard [12]	None
$\Omega_{\mu\nu} = F_{\mu\nu}$ on admissible bundle	Structural	Definition 1
Horizontal–vertical decoupling	Theorem-level (Lemma 1)	Minimal coupling
$\delta_{\mathcal{A}} S_\Pi \to D_\mu F^{\mu\nu}$	Theorem-level (Theorem 1)	None
$g_{\mathrm{YM}}^{-2} \sim \log(\Lambda/\mu)$	Scheme-dependent normalisation	UV regularization
$G_N^{-1} \sim \ell_{\mathrm{sp}}^{-2}$ vs $g_{\mathrm{YM}}^{-2} \sim \log \Lambda$	Structural hierarchy	Seeley–DeWitt
$G_\Pi = \mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$	Input from [5, 8]	Proved; $[H\text{-color}]_{\mathrm{eff}}$ in [7, 8]

7.2 Analytical status of [H-color]

The present paper takes G_Π as an input from the spectral admissibility sub-programme [5, 8]. The SU(3) sector of G_Π is established unconditionally for the colour-adapted Cayley graph $\mathrm{Cay}(G_q, S_q^{(3)})$ in [8]. For the standard Cayley graph, the effective co-admissibility $[H\text{-color}]_{\mathrm{eff}}$ — equality of the capacity exponents $\delta_c = \delta_{\omega c} = \delta_{\omega^2 c}$ in the large-q limit — is now established analytically in [7, 8] via two complementary mechanisms: the intra-coset Vandermonde symmetry (Schur's lemma for the Heisenberg centre, O31) and the modulation-frequency argument for the O12 block-averaged fingerprint observable (O32). The Theorem 1 of the present paper holds for any compact gauge group G_Π and is independent of these results; they bear only on the identification of the specific group $G_\Pi = \mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$.

The complete programme-level status is:

- $G_\Pi = \mathrm{SU}(2) \times \mathrm{U}(1)$: unconditional [5];
- $G_\Pi = \mathrm{SU}(3)$: unconditional for the colour-adapted graph; unconditional at the effective level ($[H\text{-color}]_{\mathrm{eff}}$ proved) for the standard graph [7, 8];
- The finite-q residual variance R_{var} is characterised as a modulation bias of order $O(q^{-1})$, not an open hypothesis [7].

7.3 The SU(3) Yang–Mills uniqueness problem

The open problem identified in [8] (§9.2) is the analogue of the Q6a Yang–Mills derivation for SU(3): showing that the SU(3) gauge dynamics (structure constants, covariant derivative, Yang–Mills equations) emerge uniquely from the colour triplet co-admissibility structure. This is distinct from the result of the present paper, which derives the Yang–Mills equations for any compact G_Π from the a_4 heat-kernel coefficient. The SU(3) Yang–Mills uniqueness problem concerns the *internal* structure of the SU(3) sector, not the spectral derivation of the Yang–Mills action.

7.4 What Q13 will add

The present paper establishes the two independent sectors:

$$\frac{\delta S_\Pi}{\delta g^{\mu\nu}} = c_{\rm EH}\, G_{\mu\nu} + \cdots \qquad \text{(Gravity 3.0, [6], } a_2\text{)}$$
$$\frac{\delta S_\Pi}{\delta \mathcal{A}^a_\nu} = c_{\rm YM}\, D_\mu F^{a\mu\nu} + \cdots \qquad \text{(present paper, } a_4\text{)}$$

The joint variational problem $\delta_{g,\mathcal{A}} S_\Pi = 0$, the joint Einstein–Yang–Mills equations of motion, the gauge-gravity coupling at order a_6, and the ratio $G_N/g^2_{\rm YM}$ as a precise quantitative prediction, are deferred to the companion paper Q13 (Gauge–Gravity Spectral Synthesis).

7.5 Spectral stratification as an organisational principle

Remark. The spectral stratification identified in this paper — gravity at a_2, gauge at a_4 — provides an organisational principle that differs fundamentally from purely group-theoretic unification schemes. In the latter, gravity and gauge interactions are sought at the same algebraic level, typically as different representations of a common group $G_{\rm unif}$. The present framework suggests instead that they are separated by a geometric direction (horizontal versus vertical variation on the admissible bundle) and by spectral order (two versus four derivatives in the heat-kernel expansion). Their distinct UV behaviours — quadratic divergence for gravity, logarithmic divergence for gauge — are then consequences of this geometric separation rather than independent empirical inputs. In this reading, the difficulty of incorporating gravity into the gauge-theoretic framework of the Standard Model may reflect not a missing symmetry but a difference of spectral order: gravity and gauge dynamics are not at the same level of the Seeley–DeWitt hierarchy of the admissible operator. This observation does not imply that strictly algebraic unification schemes are incorrect; it suggests that the natural organisational space for unification in the present framework is spectral and variational rather than primarily group-theoretic.

8 Conclusion

The main result of this paper is the derivation of the Yang–Mills equations $D_\mu F^{a\mu\nu} = 0$ as the vertical a_4 response of the projective spectral entropy functional $S_\Pi[g, \mathcal{A}]$, in exact parallel with the derivation of the Einstein equations as the horizontal a_2 response established in [6].

The conceptually central observation is the following. The Seeley–DeWitt degree of a coefficient encodes the physical type of the coupling it induces:

$$\begin{array}{llll} a_2 \longrightarrow & \text{gravity,} & \text{quadratic UV divergence,} & \text{perturbatively non-renormalizable sector,} \\ a_4 \longrightarrow & \text{gauge,} & \text{logarithmic UV divergence,} & \text{renormalizable sector.} \end{array}$$

As a consequence, the gauge–gravity hierarchy $G_N g^2_{\rm YM} \ll 1$ is not an empirical input but a structural output of the Seeley–DeWitt expansion: it reflects the fact that gravity and gauge dynamics live at different levels of the spectral hierarchy of the same functional $S_\Pi[g, \mathcal{A}]$.

Both sectors emerge from the same spectral cutoff $\ell_{\rm sp}$ and the same non-injective projection principle. The gauge group G_Π is inherited from the admissibility sub-programme [5, 8];

the SU(3) sector is unconditional at the effective level, with $[H\text{-color}]_{\text{eff}}$ established in [7, 8]. The spectral stratification identified here — developed in Section 7.5 — suggests more broadly that the organisational space for unification may be spectral and variational rather than primarily group-theoretic.

The complete synthesis — joint equations of motion, Born–Infeld gauge-gravity completion, and the quantitative hierarchy ratio — is addressed in Q13.

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